AS · Unit 1 · WPH11 Mechanics & Materials

Forces & Momentum

Motion tells you *how* things move. Forces tell you *why*. On this page you'll learn to draw free-body diagrams, use Newton's three laws, see why a skydiver stops speeding up, and use momentum to predict what happens when things collide or fly apart. Nearly every mechanics question starts with a free-body diagram, so that's where we begin.

Forces and free-body diagrams

A **force** is a push or a pull. It is a **vector**, so it has a size (measured in **newtons**, N) and a direction. Forces come from interactions between objects, and in mechanics you will keep meeting the same handful:

Force	Where it comes from	Direction
Weight $W = mg$	the Earth's gravitational pull on the object	vertically down, acting at the centre of gravity
Normal reaction R	a surface pushing back on an object resting on it	perpendicular to the surface, away from it
Friction	two surfaces rubbing or trying to slide over each other	along the surface, opposing the (attempted) sliding
Tension T	a stretched string, rope or cable	along the string, pulling away from the object
Drag (air resistance)	moving through a fluid; it grows with speed	opposite to the velocity
Upthrust	a fluid pushing up on anything in it (see Density, Upthrust & Viscous Drag)	vertically up

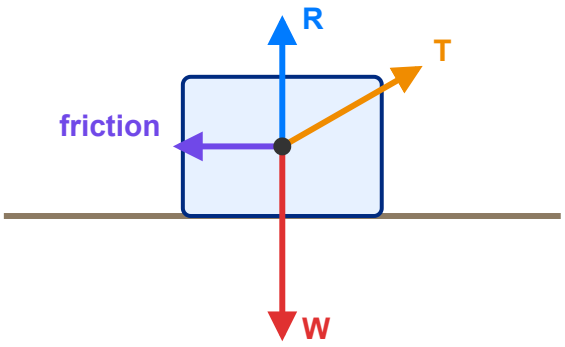
A **free-body diagram** shows **one** object on its own, with an arrow for every force acting **on** it. The rules:

- Draw only the body you are interested in, not the things around it.
- Include only forces acting **on** that body, never forces it exerts on other things.
- Each arrow points the way the force acts, with length roughly showing its size. Label every arrow.
- Don't draw the resultant force or "the force of motion". There is no such force; the resultant is what you *calculate* from the diagram.

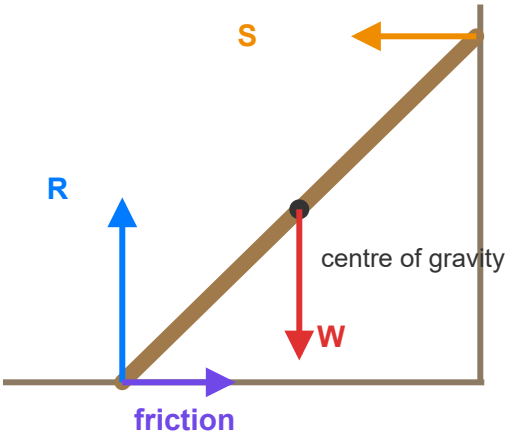
Particles and extended bodies

If you only care about how an object moves as a whole (not whether it spins), you can model it as a **particle** and draw every force from one point. If the object could rotate, like a ladder or a beam, it is an **extended body**: each force must be drawn where it actually acts, and the weight acts at the **centre of gravity**. Where a force acts matters for turning, which is the whole of the Moments page.

Particle model
box dragged by a rope



Extended body
ladder leaning on a smooth wall



Left: the box won't rotate, so every force can be drawn from one point. Right: the ladder can rotate, so each force is drawn where it acts, with the weight at the centre of gravity. The smooth wall can only push at right angles to itself, so friction at the foot stops the ladder sliding out.

DEFINITION

Centre of gravity

The **centre of gravity** of an object is the single point through which its whole weight can be taken to act. For a uniform object it is at the geometric centre.

Forces on a slope

On a slope, the trick is to **resolve along and perpendicular to the slope**, because that is the direction the object can move. The weight splits into two components (exactly as on the Motion page's vectors section):

$$\text{down the slope: } mg\sin\theta \qquad \text{into the slope: } mg\cos\theta$$

Nothing accelerates into or out of the surface, so perpendicular to the slope the forces balance and $R = mg\cos\theta$ (as long as no other force has a perpendicular component). Along the slope, whatever is left over is the resultant force that accelerates the object.

EXAM TIP

Check your components with an extreme case. At $\theta = 0$ (flat ground) nothing pulls the block along, and $\sin 0 = 0$. At $\theta = 90^\circ$ (a vertical wall) the whole weight acts along the "slope", and $\sin 90^\circ = 1$. If your expression doesn't behave like that, you've swapped sin and cos.

Newton's first and second laws

DEFINITION

Newton's first law

An object remains at rest, or continues to move with constant velocity, unless a resultant force acts on it.

This sounds obvious until you notice what it rules out. You do **not** need a force to keep something moving; you need a force to **change** its velocity. A hockey puck on smooth ice glides on and on. A car on the motorway at a steady 30 m s^{-1} still needs its engine only because the driving force has to cancel air resistance and friction. When all the forces on an object cancel, we say it is in **equilibrium**, and it is either at rest or moving at constant velocity. Those two cases look different to us, but to Newton they are the same.

When the forces don't cancel, the **resultant force** makes the object accelerate. Experiments show the acceleration is proportional to the resultant force (push twice as hard, get twice the acceleration) and inversely proportional to the mass (twice the mass, half the acceleration). Mass measures **inertia**: how strongly an object resists a change in its motion.

Newton's second law

FORMULA SHEET

$$F = ma$$

F: resultant force / N m: mass / kg a: acceleration / m s^{-2}

The newton is defined by this equation: **1 N is the resultant force that gives a mass of 1 kg an acceleration of 1 m s^{-2}**, so $1\ \text{N} = 1\ \text{kg m s}^{-2}$. The acceleration is always in the same direction as the resultant force. Newton's first law is just the special case $a = 0$: no resultant force, no acceleration.

COMMON MISTAKE

F in $F = ma$ is the **resultant** force, not "the force" or the biggest force. Choose a positive direction, add up the forces that way, and subtract the ones the other way. This form of the law also only works when the mass stays constant. (Rockets, which throw mass away, need the momentum form further down.)

WORKED EXAMPLE

A car of mass 1200 kg has a driving force of 3000 N. The total resistive force on it is 900 N. (a) Calculate its acceleration. (b) Assuming the forces stay constant, how long does it take to reach 20 m s^{-1} from rest?

Show solution

(a) Taking the direction of motion as positive, the resultant force is

$$F = 3000 - 900 = 2100 \text{ N} \qquad a = \frac{F}{m} = \frac{2100}{1200} = 1.75 \text{ m s}^{-2}$$

(b) The acceleration is constant, so use $v = u + at$:

$$t = \frac{v - u}{a} = \frac{20 - 0}{1.75} = 11.4 \text{ s}$$

WORKED EXAMPLE

A car of mass 1000 kg tows a trailer of mass 500 kg along a level road. The driving force is 2400 N. The resistive forces are 300 N on the car and 200 N on the trailer. Calculate (a) the acceleration and (b) the tension in the tow bar.

Show solution

(a) Treat the car and trailer as one body of mass 1500 kg. The tension is internal (the car pulls the trailer, the trailer pulls back on the car), so it cancels out:

$$a = \frac{2400 - 300 - 200}{1500} = \frac{1900}{1500} = 1.27 \text{ m s}^{-2}$$

(b) Now draw a free-body diagram of the trailer alone. Forwards: tension T. Backwards: 200 N. Apply $F = ma$ to the trailer:

$$T - 200 = 500 \times 1.267 \quad \Rightarrow \quad T = 833 \text{ N}$$

Check with the car: $2400 - 300 - 833 = 1267 \text{ N} = 1000 \times 1.267$. It works.

Lifts and bathroom scales

Bathroom scales don't measure your weight. They measure how hard they have to push up on you: the normal reaction R. Taking **up as positive**, Newton's second law for you in a lift gives

$$R - mg = ma \quad \Rightarrow \quad R = m(g + a)$$

If the lift accelerates upwards (a positive) the scales must push harder than your weight, and you feel heavier. If it accelerates downwards they push less, and you feel lighter. At constant velocity, in either direction, $a = 0$ and the scales read exactly mg. In free fall $a = -g$, so $R = 0$: you feel weightless even though gravity is still pulling on you as strongly as ever.

WORKED EXAMPLE

A 65 kg student stands on scales in a lift. Find the reading of the scales, in newtons, when the lift (a) accelerates upwards at 1.2 m s^{-2}, (b) moves up at a constant 3.0 m s^{-1}, (c) slows down at 1.2 m s^{-2} while moving up.

Show solution

Up is positive, so $R = m(g + a)$.

$$\text{(a)} \quad R = 65(9.81 + 1.2) = 716 \text{ N} \qquad \text{(b)} \quad R = 65 \times 9.81 = 638 \text{ N}$$

(c) Slowing down while moving up means the acceleration is **downwards**, so $a = -1.2 \text{ m s}^{-2}$:

$$R = 65(9.81 - 1.2) = 560 \text{ N}$$

Weight and gravitational field strength

Mass is the amount of matter in an object and its resistance to acceleration. It is a scalar, measured in kg, and it is the same everywhere. **Weight** is the gravitational force on the object. It is a vector, measured in N, and it depends on where you are. The link between them is the **gravitational field strength** g: the force per unit mass on an object in the field.

Gravitational field stre

FORMULA SHEET

$$g = \frac{F}{m}$$

g: field strength / N kg^{-1}

F: gravitational force / N

Weight

FORMULA SHEET

$$W = mg$$

W: weight / N m: mass / kg

DEFINITION

Gravitational field strength

The gravitational field strength at a point is the gravitational force per unit mass acting on a small mass placed at that point. Near the Earth's surface $g = 9.81\ \text{N kg}^{-1}$.

Now put $W = mg$ into Newton's second law for an object falling with no air resistance. The only force is its weight, so $a = \dfrac{F}{m} = \dfrac{mg}{m} = g$. That's why every object falls with the same acceleration: a heavier object is pulled harder, but it also has more inertia, and the two effects cancel exactly. It also shows that $9.81\ \text{N kg}^{-1}$ and $9.81\ \text{m s}^{-2}$ are the same thing. To measure g yourself, see Core Practical 1 on the Motion page.

WORKED EXAMPLE

An astronaut has a mass of 80 kg. On the Moon $g = 1.62\ \text{N kg}^{-1}$. (a) Find her weight on Earth and on the Moon. (b) What horizontal resultant force would give her an acceleration of 2.0 m s⁻² on the Moon?

Show solution

$$\text{(a)}\quad W_{\text{Earth}} = 80 \times 9.81 = 785\ \text{N}, \qquad W_{\text{Moon}} = 80 \times 1.62 = 130\ \text{N}$$

(b) $F = ma = 80 \times 2.0 = 160\ \text{N}$, exactly the same as on Earth. Her **weight** changed, but her **mass** (her inertia) did not. A Moon walker finds things easy to lift but just as hard to push into motion sideways.

Terminal velocity

Drop something through air or water and the drag on it grows as it speeds up. Follow a skydiver through Newton's second law, taking down as positive:

1. **Just after jumping**, $v = 0$ so there is no drag. The resultant force is the weight and $a = g$.
2. **As she speeds up**, the drag D increases, so the resultant force $W - D$ gets smaller. She is still speeding up, but the acceleration decreases: the v–t graph curves over.
3. **Eventually** the drag equals her weight. The resultant force is zero, so (Newton's first law) she carries on at a constant velocity: her **terminal velocity**.
4. **Opening the parachute** makes the drag at that speed far bigger than her weight. The resultant force is now upwards, so she **decelerates**. As she slows the drag falls, until it equals her weight again at a new, much lower terminal velocity.

DEFINITION

Terminal velocity

The constant, maximum velocity reached by an object falling through a fluid, when the drag (together with any upthrust) balances its weight, so the resultant force is zero.

EXAM TIP

"Describe and explain the motion" questions want the chain of reasoning, not just "it reaches terminal velocity". Say what happens to the drag, then the resultant force, then the acceleration, then the gradient of the v–t graph. A small ball falling through oil in Core Practical 2 does exactly the same thing, with its drag given by Stokes' law (see Density, Upthrust & Viscous Drag).

WORKED EXAMPLE

A skydiver and her kit have a total mass of 85 kg. (a) What is the drag on her at terminal velocity? (b) Just after her parachute opens, the drag is 2400 N. Calculate her acceleration at that moment.

Show solution

(a) At terminal velocity the resultant force is zero, so drag = weight:

$$D = mg = 85 \times 9.81 = 834 \text{ N}$$

(b) The resultant force is now upwards:

$$F = 2400 - 834 = 1566 \text{ N} \qquad a = \frac{1566}{85} = 18.4 \text{ m s}^{-2} \text{ upwards}$$

She is still moving **down**; the upward acceleration means she is slowing down.

Newton's third law

DEFINITION

Newton's third law

If body A exerts a force on body B, then body B exerts a force on body A that is equal in magnitude and opposite in direction.

Forces always come in pairs because they come from **interactions**. You can't push without being pushed back: lean on a wall and the wall leans on you. The two forces in a Newton's third law pair always:

- act on **two different bodies** (one on each);
- are the **same type** of force (both gravitational, both contact, both magnetic…);
- are **equal in magnitude** and **opposite in direction**, along the same line;
- exist for exactly the **same time**: they start and stop together.

Examples are everywhere. A swimmer pushes water backwards and the water pushes her forwards. A car's tyres push the road backwards and the road pushes the car forwards. The Earth pulls the Moon, and the Moon pulls the Earth just as hard (that pull on the Earth is what raises the tides).

COMMON MISTAKE

The classic wrong answer: "the weight of a book on a table and the normal reaction form a Newton's third law pair". They don't. Both act on the **same** body (the book) and they are different **types** (gravitational and contact). The partner of the book's weight is the book's gravitational pull **on the Earth**. The partner of the table's push on the book is the book's push **on the table**.

Momentum

Which is harder to stop: a tennis ball at 50 m s⁻¹ or a lorry at 1 m s⁻¹? Both mass and velocity matter, and the quantity that combines them is **momentum**.

Momentum

FORMULA SHEET

$$p = mv$$

p: momentum / kg m s⁻¹ m: mass / kg v: velocity / m s⁻¹

Momentum is a **vector**, in the same direction as the velocity, so in one-dimensional problems it carries a sign. A 2 kg trolley moving left at 3 m s⁻¹ has momentum $-6\ \text{kg m s}^{-1}$ if you chose right as positive. Its unit can also be written N s, since $1\ \text{N s} = 1\ \text{kg m s}^{-2} \times \text{s} = 1\ \text{kg m s}^{-1}$.

PREVIEW · UNIT 4

Force is the rate of change of momentum

You'll meet this properly in Unit 4 (on the Impulse & 2D Collisions page), but it's worth seeing now how Newton's second law connects to momentum.

From F = ma to F = Δp/Δt

1. Start from Newton's second law and write the acceleration as change in velocity over time:

$$F = ma = m\,\frac{v - u}{\Delta t}$$

2. If the mass is constant it can go inside the bracket:

$$F = \frac{mv - mu}{\Delta t} = \frac{\Delta p}{\Delta t}$$

*The resultant force equals the rate of change of momentum. Multiplying by Δt gives the **impulse** $F\Delta t = \Delta p$, which is why airbags and crumple zones work: making Δt longer makes the force for the same Δp smaller.*

Conservation of linear momentum

DEFINITION

Principle of conservation of linear momentum

The total linear momentum of a system of interacting bodies remains constant, provided no resultant external force acts on the system.

"External" is the important word. In a collision the bodies push on each other very hard, but those are internal forces. Newton's third law guarantees they come in equal and opposite pairs, so whatever momentum one body gains, the other loses. Here's the argument in steps.

Conservation of momentum from Newton's second and third laws

1. Two bodies A and B collide. During the collision, A pushes on B with force F. By Newton's third law, B pushes on A with force $-F$, and the two forces last for the same time Δt.

2. Newton's second law in momentum form, $F = \Delta p/\Delta t$, applied to each body:

$$\Delta p_B = F\,\Delta t \qquad\qquad \Delta p_A = -F\,\Delta t$$

3. Add them to get the change in the total momentum:

$$\Delta p_A + \Delta p_B = -F\Delta t + F\Delta t = 0$$

The total cannot change. If an outside force (friction, say) also acted, it would add its own $F\Delta t$ and the total would change, which is why the principle needs "no resultant external force".

4. So for any collision or explosion in one dimension:

$$\boxed{m_A u_A + m_B u_B = m_A v_A + m_B v_B}$$

Collisions and explosions

- **Collisions:** write the total momentum before and after, with signs, and solve for the unknown. If the bodies stick together, they share one final velocity.
- **Explosions and recoil:** if everything starts at rest, the total momentum is zero before, so it is zero after. The pieces fly apart with equal and opposite momenta: $m_A v_A = -m_B v_B$. The lighter piece goes faster. A gun recoils, a skater pushing off a wall glides backwards.

EXAM TIP

Momentum is conserved in every collision and explosion (with no external forces). **Kinetic energy usually isn't**: in a collision some is transferred to thermal energy and sound, and in an explosion extra kinetic energy comes from chemical or elastic stores. Never use "KE before = KE after" unless you are told the collision is elastic. Checking whether a collision is elastic is Unit 4 content.

WORKED EXAMPLE

A trolley of mass 0.80 kg moving at 0.50 m s^{-1} hits a stationary trolley of mass 1.20 kg. They stick together. (a) Find their velocity after the collision. (b) How much kinetic energy is transferred to other stores?

Show solution

(a) $0.80 \times 0.50 + 1.20 \times 0 = (0.80 + 1.20)\,v \quad \Rightarrow \quad v = \dfrac{0.40}{2.00} = 0.20 \text{ m s}^{-1}$

(b) $E_{k,\text{before}} = \frac{1}{2}(0.80)(0.50)^2 = 0.100 \text{ J}, \qquad E_{k,\text{after}} = \frac{1}{2}(2.00)(0.20)^2 = 0.040 \text{ J}$

So 0.060 J (60% of the kinetic energy) is transferred, mostly to thermal energy and sound.

WORKED EXAMPLE

A rifle of mass 4.0 kg fires a bullet of mass 10 g at 400 m s^{-1}. Calculate the recoil velocity of the rifle.

Show solution

Everything starts at rest, so the total momentum is zero before and after. Taking the bullet's direction as positive:

$$0 = 0.010 \times 400 + 4.0\,v \quad \Rightarrow \quad v = -\frac{4.0}{4.0} = -1.0 \text{ m s}^{-1}$$

The rifle moves back at 1.0 m s^{-1}. The minus sign tells you the direction, so keep it in your answer or state "backwards".

WORKED EXAMPLE

A railway truck of mass 12 000 kg moving at 3.0 m s^{-1} hits a stationary truck of mass 8000 kg, and they couple together. The coupled trucks then roll 45 m before stopping. Calculate the average resistive force on them.

Show solution

Step 1, the collision (momentum):

$$12\,000 \times 3.0 = 20\,000\,v \quad \Rightarrow \quad v = 1.8 \text{ m s}^{-1}$$

Step 2, the deceleration (SUVAT): with $u = 1.8$, $v = 0$, $s = 45$:

$$0 = 1.8^2 + 2a(45) \quad \Rightarrow \quad a = -\frac{3.24}{90} = -0.036 \text{ m s}^{-2}$$

Step 3, the force (Newton's second law):

$$F = ma = 20\,000 \times 0.036 = 720 \text{ N}$$

Momentum is conserved during the brief collision, but afterwards the resistive force is an external force, so the momentum of the trucks falls to zero.

SUGGESTED EXPERIMENT

Investigating conservation of momentum with an air track

Two gliders sit on an air track, which floats them on a cushion of air so friction (an external force) is almost zero. Each glider carries a card of known length that passes through a light gate before and after the collision; velocity = card length ÷ time the gate is blocked. Weigh the gliders on a balance, then compare the total momentum before and after. Velcro or a pin and putty makes them stick; repelling magnets make them bounce. Level the track carefully, or a component of weight acts along it as an external force.

Quick checks

QUICK CHECK

A book rests on a table. Which force forms a Newton's third law pair with the weight of the book?

- The normal contact force of the table on the book
- The gravitational pull of the book on the Earth
- The normal contact force of the book on the table
- The weight of the table

The weight is the Earth pulling the book gravitationally, so its partner must be the book pulling the Earth gravitationally: different body, same type, equal and opposite. The table's push on the book acts on the same body as the weight, so it can't be the partner.

QUICK CHECK

A car travels along a straight, level road at a constant 25 m s^{-1}. Which statement is correct?

- The driving force is greater than the resistive forces.
- The resultant force on the car is zero.
- There are no forces acting on the car.
- The resultant force acts in the direction of motion.

Constant velocity means zero acceleration, so by Newton's first law the resultant force is zero. Plenty of forces act; they just cancel, with the driving force equal to the resistive forces.

QUICK CHECK

A lift is moving upwards and slowing down. Compared with the passenger's weight, the reading on bathroom scales in the lift is:

- greater
- less
- equal
- zero

Slowing down while going up means the acceleration is downwards, so a is negative in $R = m(g + a)$ and $R < mg$. The direction of the *velocity* doesn't matter; only the acceleration does.

QUICK CHECK

A skydiver falling at terminal velocity opens her parachute. Immediately afterwards:

- her velocity is upwards
- the resultant force on her is upwards, so she slows down
- the drag equals her weight
- her acceleration is 9.81 m s^{-2} downwards

The drag suddenly exceeds her weight, so the resultant force and the acceleration point up. She is still moving down, just more slowly each second, until the drag falls back to equal her weight.

QUICK CHECK

A stationary firework explodes into two pieces. Straight after the explosion, which statement is true?

- The total momentum is zero and the total kinetic energy is zero.
- The total momentum is zero and the total kinetic energy has increased.
- The total momentum and the total kinetic energy have both increased.
- Momentum is not conserved, because energy was released.

Momentum is conserved, so the total stays at zero: the pieces have equal and opposite momenta. Kinetic energy is a scalar and can't cancel; it comes from the chemical energy store of the explosive.

QUICK CHECK

A 2.0 kg trolley moving at 3.0 m s⁻¹ to the right hits a 1.0 kg trolley moving at 4.0 m s⁻¹ to the left. They stick together. What is their velocity afterwards?

- 0.67 m s⁻¹ to the right
- 3.3 m s⁻¹ to the right
- 0.67 m s⁻¹ to the left
- zero

With right as positive: $2.0 \times 3.0 + 1.0 \times (-4.0) = 2.0 \text{ kg m s}^{-1}$, and $v = 2.0/3.0 = 0.67 \text{ m s}^{-1}$. Forgetting the minus sign gives 3.3 m s⁻¹.